Seed Fund Budget Allocation Table

Agent-Based Vision Mixer for Video Broadcasts

Auto Surveillance of Vehicles (ASV)

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Supervisor: Dr. Omar Osman Khan
Institution: FAST National University of Computer and Emerging Sciences (FAST NUCES)
Program: Aspire Institute Seed Fund Round 1, 2026 **Award Requested:** $500 USD
Project Duration: 9 Weeks (Research & Prototype Phase)

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1. Budget Overview

This project is an academic research and software development initiative. The system is implemented entirely in Python using open-source libraries; therefore, the primary cost drivers are cloud compute access for model training/evaluation, datasets, software subscriptions, and dissemination activities. No hardware procurement or fieldwork is required. All seed funding will be directed toward activities that directly accelerate the 9-week prototype delivery.

The total seed funding requested is **$500 USD**. The budget is structured across six categories: Cloud Compute & GPU Access, Dataset & Benchmark Licensing, Development Tools & Software, Testing & Evaluation Infrastructure, Dissemination & Documentation, and IEEE Research Access.

2. Detailed Budget Allocation

Category / Item	Justification	Unit Cost (USD)	Qty	Subtotal (USD)
Category 1: Cloud Compute & GPU Access				
Google Colab Pro+ Subscription	Required to run YOLOv8 and ResNet-IBN inference on GPU (T4/A100). Local machines lack CUDA-capable hardware for real-time multi-stream processing.	$12/mo	3 mo	$36
AWS EC2 GPU Instance (g4dn.xlarge)	Burst compute for full regression testing of all 20 test cases across 5–10 simultaneous RTSP streams. Estimated 20 hrs of usage across Weeks 7–8.	$0.526/hr	40 hr	$22

Category / Item	**Justification**	**Unit Cost (USD)**	**Qty**	**Subtotal (USD)**
Cloud Storage (AWS S3 / Google Drive)	Storing benchmark video clips (CityFlow subset), model weights, and intermediate video stitching outputs during Weeks 4–8.	$2/mo	3 mo	$6
			Category 1 Subtotal	**$64**
Category 2: Dataset & Benchmark Access				
CityFlow / AI City Challenge Dataset Access	Official benchmark dataset for multi-camera vehicle re-identification evaluation. Required for TC-RI-01 through TC-RI-05 and ReID accuracy baseline. Public download, no cost.	$0	1	$0
VehicleX / VERI-776 Subset Download	Pre-training dataset for ResNet-IBN vehicle Re-ID model. 50K+ vehicle images across 776 identities.	$5	1	$5
Locally Recorded Test Clips (Storage Media)	USB storage for 3–10 simulated RTSP streams recorded locally using dashcam / webcam to supplement public datasets for edge-case testing.	$15	2	$30
RTSP Stream Simulation License (iSpy / Agent DVR)	Software to simulate RTSP streams from pre-recorded MP4 files for multi-camera testing without physical IP cameras.	$10	1	$10
			Category 2 Subtotal	**$45**
Category 3: Development Tools & Software				
GitHub Copilot (Individual, 2 Accounts)	AI-assisted code generation to accelerate module-by-module development across Weeks 2–7. Reduces boilerplate in DeepSORT integration, FFmpeg subprocess calls, and SMTP wrappers.	$10/mo	3 mo	$60
PyCharm Professional License (Student)	Advanced Python IDE with profiling, database inspection, and remote SSH capabilities needed for debugging multi-threaded video pipeline. (Student pricing.)	$0	1	$0

Category / Item	**Justification**	**Unit Cost (USD)**	**Qty**	**Subtotal (USD)**
Weights & Biases (wandb) Team Plan	Experiment tracking for YOLOv8 fine-tuning hyperparameters and ReID cosine similarity threshold optimization.	$0	1	$0
Overleaf Professional Subscription	Required for collaborative LaTeX writing of the final FYP report, conference paper draft, and this application document.	$8/mo	2 mo	$16
FFmpeg (Open Source)	Video stitching and H.264 encoding library. No cost.	$0	1	$0
SQLite / PostgreSQL (Open Source)	Database for clip metadata, vehicle feature embeddings, and tracking records. No cost.	$0	1	$0
			Category 3 Subtotal	**$76**
Category 4: Testing & Evaluation Infrastructure				
Webcams / USB Cameras for Simulated Streams	Two additional USB webcams to serve as physical camera inputs for live end-to-end demo (Week 7). Minimum 720p resolution.	$30	2	$60
USB-C Hub / Video Capture Cards	Connect multiple USB cameras to a single development machine simultaneously. Required for multi-stream pipeline testing.	$25	1	$25
SMTP Test Account (SendGrid / Mailgun)	Programmatic email testing for TC-VE-03 and TC-VE-05. Free tier sufficient; no cost.	$0	1	$0
Printouts: Test Case Execution Reports	Physical printed copies of the 20-test-case execution report for supervisor review and submission.	$1	20 pg	$20
			Category 4 Subtotal	**$105**
Category 5: Dissemination & Documentation				
arXiv Preprint Submission	No cost. Project outputs will be submitted as an open-access preprint to arXiv (cs.CV).	$0	1	$0
Conference Paper Registration Fee (Target: IEEE ICIP / Workshop)	Workshop submission fee for the methodology paper covering the DBN camera switching engine.	$50	1	$50

Category / Item	**Justification**	**Unit Cost (USD)**	**Qty**	**Subtotal (USD)**
FYP Report Binding & Printing (Final Submission)	Hardbound final project report for university library and supervisor submission. Printed color copies of diagrams and test results.	$20	2 copies	$40
Demonstration Video Production	Screen-recording software to produce the final demonstration video. DaVinci Resolve (free).	$0	1	$0
Poster Printing (Aspire Demo Day / FYP Expo)	A1 size poster for project exhibition. Printed at campus print center.	$10	1	$10
Contingency Reserve	5% contingency for unforeseen compute overruns, additional storage, or unexpected software requirements.	Flat		$10
			Category 5 Subtotal	**$110**
Category 6: IEEE Research Access				
IEEE Xplore Digital Library Access	One-month institutional-equivalent access to IEEE Xplore for downloading relevant papers on multi-camera vehicle tracking, Re-ID benchmarks, and camera switching algorithms. Required to perform a thorough literature review and ensure novelty of the proposed approach.	$30/mo	1 mo	$30
IEEE Paper Submission (Author Processing Charge)	Open-access author fee for submitting the final research paper to an IEEE journal or conference proceedings (e.g., IEEE TITS or IEEE ICIP). Budgeted conservatively; actual fee depends on venue.	$70	1	$70
			Category 6 Subtotal	**$100**
			TOTAL SEED FUNDING REQUESTED	**$500**

3. Budget Summary by Category

Category	Allocated (USD)	% of Total
Cloud Compute & GPU Access	$64	12.8%
Dataset & Benchmark Access	$45	9.0%
Development Tools & Software	$76	15.2%
Testing & Evaluation Infrastructure	$105	21.0%
Dissemination & Documentation	$110	22.0%
IEEE Research Access	$100	20.0%
Total Requested	**$500**	**100%**

4. Budget-to-Timeline Alignment

The table below maps seed fund expenditure to the 9-week project timeline.

Week	Phase	Primary Expenditure	Est. Spend (USD)
W1	Environment Setup	USB webcams, Colab Pro+, RTSP simulator, Overleaf, IEEE Xplore access	$112
W2	Vehicle Detection	Cloud compute (Colab), dataset download, storage	$20
W3	Single-Camera Tracking	Cloud compute, Copilot subscription	$15
W4	Cross-Camera Re-ID	Cloud compute (EC2 burst), VERI-776	$20
W5	Camera Switching Engine	Cloud compute, test clip storage	$15
W6	Video Stitching & Email	Cloud compute, SMTP test account	$15
W7	Integration & Live Demo	USB hub, capture cards, multi-stream testing	$40
W8	System Testing	EC2 burst (full regression), printouts	$30
W9	Documentation & Dissemination	Report binding, poster, conference fee, IEEE paper submission	$133
—	**Total**		**$400**
—	**Contingency**	(Available within $500 cap)	**$100**

5. Notes on Cost Efficiency

1. **Open-source first:** All core AI/ML frameworks (PyTorch, Ultralytics, DeepSORT, FFmpeg, smtplib, SQLite) are free and open-source. Seed funding targets only resources not freely available.

2. **Institutional leverage:** FAST NUCES lab GPU access and GitHub Student Pack significantly reduce compute and tooling costs.

3. **No fieldwork costs:** The system uses publicly available benchmark datasets (CityFlow, AI City Challenge) and simulated RTSP streams. No travel or field study costs.

4. **No hardware procurement:** The system is a pure software pipeline. Physical cameras in the budget are low-cost USB webcams used only for the live demonstration.

5. **IEEE access is essential:** IEEE Xplore access is required to conduct a rigorous literature review for the research paper. The author processing charge enables open-access publication, maximizing visibility and impact of the research.

6. **Conservative conference fee:** The $50 conference registration is a placeholder; if a free workshop is identified, this will be reallocated to compute or documentation.

7. **Tight timeline manageability:** The modular development approach (one module per week, W2–W6) ensures the budget is spent progressively and reviewable at each milestone.

Budget prepared by: Sami Ullah & Khushal Das | Supervised by: Dr. Omar Osman Khan | FAST NUCES, Department of Computer Science | April 2026